

MTH 490: Equivalence and Absolute Value

Problems 16, 19, 20, 21

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1.3 Order and Absolute Value

Although the real numbers satisfy the field axioms given in the previous section, these algebraic properties don't fully characterize $\mathbb{R}$; one thing that's missing is the notion of *order*. We begin by introducing the concept of positive and negative numbers.

Axiom 1.4 (Order). There exists a subset of the real numbers called the set of **positive numbers**, denoted $\mathbb{R}^+$, that satisfies the following two conditions:

(a) If $a, b \in \mathbb{R}^+$ then

$$a + b \in \mathbb{R}^+ \quad \text{and} \quad a \cdot b \in \mathbb{R}^+.$$

In other words, the positive numbers are closed under addition and multiplication.

(b) Given any real number a , one and only one of the following is true:

- (i) a is positive.
- (ii) a is zero.
- (iii) $-a$ is positive.

The complement of the set $\mathbb{R}^+ \cup \{0\}$, i.e. the set of numbers a that aren't positive or zero, is called the set of **negative numbers** and is denoted $\mathbb{R}^-$.

Definition 1.5. We define an order relation on $\mathbb{R}$ as follows: for all real numbers a and b we say that a is **less than** b ($a < b$) if $b - a \in \mathbb{R}^+$. In other words, $a < b$ means that the number $b - a$ is positive. For all real numbers, a and b , we say that $a \leq b$ if $b - a \in \mathbb{R}^+ \cup \{0\}$. In other words, $a \leq b$ means that the number $b - a$ is positive or zero. The symbols $>$ and $\geq$ are defined similarly.

16. Given real numbers $a, b \geq 0$ show that $a^2 \leq b^2$ if and only if $a \leq b$. Show that a similar result holds if the inequality is strict.

Proof. First, if $x, y \in \mathbb{R}^+ \cup \{0\}$ then $xy \in \mathbb{R}^+ \cup \{0\}$: if $x = 0$ or $y = 0$ then $xy = 0$, and otherwise $xy \in \mathbb{R}^+$ by Axiom 1.4(a).

($\Leftarrow$) Suppose $a \leq b$. By Definition 1.5, $b - a \in \mathbb{R}^+ \cup \{0\}$. Since $a, b \geq 0$, we have $b + a \in \mathbb{R}^+ \cup \{0\}$. By the field axioms,

$$b^2 - a^2 = (b - a)(b + a) \in \mathbb{R}^+ \cup \{0\},$$

so $a^2 \leq b^2$.

($\Rightarrow$) Suppose $a^2 \leq b^2$, and assume for contradiction that $a \leq b$ is false. Then $b - a \notin \mathbb{R}^+ \cup \{0\}$, so by Axiom 1.4(b), $-(b - a) = a - b \in \mathbb{R}^+$; that is, $b < a$.

Since $a = (a - b) + b$ with $a - b \in \mathbb{R}^+$ and $b \in \mathbb{R}^+ \cup \{0\}$, we get $a \in \mathbb{R}^+$, and then $a + b \in \mathbb{R}^+$. So

$$a^2 - b^2 = (a - b)(a + b) \in \mathbb{R}^+$$

by Axiom 1.4(a). But $a^2 \leq b^2$ means $-(a^2 - b^2) = b^2 - a^2 \in \mathbb{R}^+ \cup \{0\}$. Then both $a^2 - b^2$ and $-(a^2 - b^2)$ are positive or zero, contradicting the “one and only one” condition of Axiom 1.4(b). Hence $a \leq b$.

Strict version: $a^2 < b^2$ if and only if $a < b$.

($\Leftarrow$) Suppose $a < b$. Then $b - a \in \mathbb{R}^+$, and $b = (b - a) + a$ with $a \in \mathbb{R}^+ \cup \{0\}$, so $b \in \mathbb{R}^+$ and $b + a \in \mathbb{R}^+$. Therefore $b^2 - a^2 = (b - a)(b + a) \in \mathbb{R}^+$ by Axiom 1.4(a), so $a^2 < b^2$.

($\Rightarrow$) Suppose $a^2 < b^2$, and assume for contradiction that $a < b$ is false. Then $b - a \notin \mathbb{R}^+$, so by Axiom 1.4(b) either $b - a = 0$ or $a - b \in \mathbb{R}^+$. In either case $a - b \in \mathbb{R}^+ \cup \{0\}$, so $b \leq a$. By the non-strict result (with a and b swapped), $b^2 \leq a^2$, so $a^2 - b^2 \in \mathbb{R}^+ \cup \{0\}$. But $a^2 < b^2$ means $-(a^2 - b^2) \in \mathbb{R}^+$, contradicting Axiom 1.4(b). Hence $a < b$. $\square$

Definition 1.6. Given two distinct real numbers a and b such that $a < b$, we denote the set $\{x \in \mathbb{R} : a < x < b\}$ with the notation (a, b) . We denote $\{x \in \mathbb{R} : a \leq x \leq b\}$ with the notation $[a, b]$. In words (a, b) is the set of all real numbers between a and b . The set (a, b) is called an **open interval**. The set $[a, b]$ is called a **closed interval**. The values a and b are called the **endpoints**.

We will generalize the notation for intervals in the usual ways. For instance, $[a, b)$ is understood to mean the set $\{x \in \mathbb{R} : a \leq x < b\}$. We may also replace a with $-\infty$ or b with ∞ in which case our interval is **unbounded**. If both a and b are finite, the interval is **bounded**.

Definition 1.7. The **absolute value** of a real number x is denoted by $|x|$ and defined as

$$|x| = \begin{cases} x & \text{if } x \geq 0 \\ -x & \text{if } x < 0 \end{cases}.$$

19. Show that for all $a, b \in \mathbb{R}$,

(a) $|a \cdot b| = |a| \cdot |b|$

(b) $|a + b| \leq |a| + |b|$

Proof of (a). By Definition 1.7, there are two cases for $|a|$ and two cases for $|b|$, giving four cases. We use the fact from the proof of Problem 16 that the product of two elements of $\mathbb{R}^+ \cup \{0\}$ is in $\mathbb{R}^+ \cup \{0\}$.

Case 1: $a \geq 0, b \geq 0$. Then $ab \in \mathbb{R}^+ \cup \{0\}$, so $|ab| = ab$. Also $|a| = a$ and $|b| = b$, so

$$|a| \cdot |b| = a \cdot b = |a \cdot b|.$$

Case 2: $a < 0, b \geq 0$. Then $|a| = -a$ and $|b| = b$, so $|a| \cdot |b| = (-a)b = -(ab)$. Since $-a \in \mathbb{R}^+$ and $b \in \mathbb{R}^+ \cup \{0\}$, we have $-(ab) = (-a)b \in \mathbb{R}^+ \cup \{0\}$. If $-(ab) \in \mathbb{R}^+$, then $ab < 0$ and $|ab| = -(ab)$ by Definition 1.7. If $-(ab) = 0$, then $ab = 0$ and $|ab| = 0 = -(ab)$. Either way,

$$|a \cdot b| = -(ab) = |a| \cdot |b|.$$

Case 3: $a \geq 0, b < 0$. Then $|a| \cdot |b| = a(-b) = -(ab)$, and the argument of Case 2 with the roles of a and b exchanged gives $|a \cdot b| = -(ab) = a \cdot (-b) = |a| \cdot |b|$.

Case 4: $a < 0, b < 0$. Then $|a| = -a$ and $|b| = -b$, and by the field axioms

$$|a| \cdot |b| = (-a)(-b) = (-1)a \cdot (-1)b = (-1)^2 ab = ab.$$

Since $-a, -b \in \mathbb{R}^+$, Axiom 1.4(a) gives $ab = (-a)(-b) \in \mathbb{R}^+$, so $|ab| = ab$. Hence $|a \cdot b| = |a| \cdot |b|$.

In all four cases, $|a \cdot b| = |a| \cdot |b|$. $\square$

Proof of (b). We use two facts. First, adding the same number to both sides preserves order: for $p, q, r \in \mathbb{R}$, $(q + r) - (p + r) = q - p$ by the field axioms, so $p \leq q$ implies $p + r \leq q + r$. Second, if $x < 0$ then $x < -x$, since $(-x) - x = (-x) + (-x) \in \mathbb{R}^+$ by Axiom 1.4(a); and if $x \geq 0$ then $-x \leq x$, since $x - (-x) = x + x \in \mathbb{R}^+ \cup \{0\}$. In words, a negative number is less than a positive one.

By Definition 1.7 there are again four cases.

Case 1: $a \geq 0, b \geq 0$. Then $a + b \in \mathbb{R}^+ \cup \{0\}$, so

$$|a + b| = a + b = |a| + |b|.$$

Cases 2 and 3: one of a, b is negative and the other is nonnegative. Since $a + b = b + a$ and $|a| + |b| = |b| + |a|$, the statement is symmetric in a and b , so without loss of generality let $a < 0$ and $b \geq 0$. Then

$$|a| + |b| = -a + b, \quad |a + b| = \begin{cases} a + b & \text{if } a + b \geq 0 \\ -(a + b) & \text{if } a + b < 0 \end{cases}.$$

If $a + b \geq 0$: since $a < 0$, we have $a < -a$, and adding b gives $a + b \leq -a + b$. So $|a + b| \leq |a| + |b|$.

If $a + b < 0$: since $b \geq 0$, we have $-b \leq b$, and adding $-a$ gives $-a + (-b) \leq -a + b$. So $|a + b| = -(a + b) = -a + (-b) \leq |a| + |b|$.

Case 4: $a < 0, b < 0$. Then $-a, -b \in \mathbb{R}^+$, so $-(a + b) = (-a) + (-b) \in \mathbb{R}^+$ by Axiom 1.4(a), meaning $a + b < 0$. Therefore

$$|a + b| = -(a + b) = -a + (-b) = |a| + |b|.$$

In all four cases, $|a + b| \leq |a| + |b|$. $\square$

20. Show that for $a, b, c \in \mathbb{R}$,

(a) $|a - b| \geq ||a| - |b||$

(b) $|a - b| \leq |a - c| + |b - c|$

Proof of (a). By Problem 19(b),

$$|a| = |(a - b) + b| \leq |a - b| + |b|, \quad \text{so} \quad |a| - |b| \leq |a - b|.$$

Swapping the roles of a and b gives $|b| - |a| \leq |b - a|$. By Problem 17(d), $|b - a| = |-(a - b)| = |a - b|$, so $-(|a| - |b|) \leq |a - b|$.

Let $x = |a| - |b|$. We have shown $x \leq |a - b|$ and $-x \leq |a - b|$. If $x \geq 0$, then $|x| = x \leq |a - b|$. If $x < 0$, then $|x| = -x \leq |a - b|$. In both cases, by Definition 1.7, $||a| - |b|| \leq |a - b|$. $\square$

Proof of (b). By the field axioms, $a - b = (a - c) + (c - b)$. Applying Problem 19(b) to the real numbers $a - c$ and $c - b$,

$$|a - b| = |(a - c) + (c - b)| \leq |a - c| + |c - b|.$$

By Problem 17(d), $|c - b| = |-(b - c)| = |b - c|$. Therefore

$$|a - b| \leq |a - c| + |b - c|. \quad \square$$

21. Suppose that $x, a \in \mathbb{R}$ and $\epsilon > 0$ then

(a) $|x| < \epsilon$ if and only if $-\epsilon < x < \epsilon$

(b) $|x - a| < \epsilon$ if and only if $a - \epsilon < x < a + \epsilon$.

Proof of (a). We use Problem 11: if $p < q$ then $-q < -p$. We also use that if $p < q$ and $q \leq r$, or if $p \leq q$ and $q < r$, then $p < r$: we have $r - p = (r - q) + (q - p)$ where one term is in $\mathbb{R}^+$ and the other is in $\mathbb{R}^+ \cup \{0\}$, so $r - p \in \mathbb{R}^+$ (directly if one term is 0, and by Axiom 1.4(a) otherwise).

By Definition 1.7, each direction splits into two cases according to the sign of x .

($\Rightarrow$) Suppose $|x| < \epsilon$.

Case 1: $x < 0$. Then $|x| = -x$, so $-x < \epsilon$. By Problem 11, $-\epsilon < -(-x) = x$. Also $x < 0$ and $0 < \epsilon$, so $x < \epsilon$. Hence $-\epsilon < x < \epsilon$.

Case 2: $x \geq 0$. Then $|x| = x$, so $x < \epsilon$. Since $0 < \epsilon$, Problem 11 gives $-\epsilon < 0$, and with $0 \leq x$ this gives $-\epsilon < x$. Hence $-\epsilon < x < \epsilon$.

($\Leftarrow$) Suppose $-\epsilon < x < \epsilon$.

Case 1: $x \geq 0$. Then $|x| = x$, and $x < \epsilon$ is given, so $|x| < \epsilon$.

Case 2: $x < 0$. Then $|x| = -x$. From $-\epsilon < x$, Problem 11 gives $-x < -(-\epsilon) = \epsilon$, so $|x| < \epsilon$.

In every case the two statements are equivalent, so $|x| < \epsilon$ if and only if $-\epsilon < x < \epsilon$. $\square$

Proof of (b). Applying part (a) to the real number $x - a$,

$$|x - a| < \epsilon \iff -\epsilon < x - a < \epsilon.$$

For any $p, q \in \mathbb{R}$, $(q + a) - (p + a) = q - p$ by the field axioms, so $p < q$ if and only if $p + a < q + a$. Adding a to each part of the inequality gives

$$-\epsilon < x - a < \epsilon \iff a - \epsilon < x < a + \epsilon. \quad \square$$